INCH-POUND

CCC-D-950F
INTERIM AMENDMENT 1
October 31, 1990

INTERIM AMENDMENT

TO

FEDERAL SPECIFICATION

DYEING AND AFTERTREATING PROCESSES FOR COTTON CLOTHS

This interim amendment was developed by the U.S. Army Natick Research, Development and Engineering Center, Natick, MA 01760-5014 based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this interim amendment as a valid exception to CCC-D-950F, dated June 26, 1989.

PAGE 2

2.1, under Federal Standards: Add the following:

"FED-STD-595 - Colors Used in Government Procurement"

PAGE 5

Add the following new paragraph and tables:

"3.6.2 Spectral reflectance. When olive green shades or olive drab shades are specified, the spectral reflectance factors (in percent) for visible/near infrared wavelength range of 600 to 860 nanometers (nm) of the finished cloth shall meet the requirements specified in table AI. When Tan 380 or Tan 686 matching color chip 33446 of FED-STD-595 is specified, the spectral reflectance factors (in percent) for visible/near infrared wavelength range of 700 to 860 nm of the finished cloth shall meet the requirements specified in table AII. Testing shall be as specified in 4.4.3.

AMSC N/A FSC 8305

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:31Z
Check the source to verify that this is the current version before use.

TABLE AI. Spectral reflectance values (percent) for olive green or olive drab shades

Wavelength, Nanometers (nm)	Reflectance Min.	Max.	Wavelength Nanometers (nm)	Reflectance Min.	Max.
600	3	12	740	7	52
620	3	12	760	11	60
640	3	12	780	17	64
660	3	13	800	24	70
680	3	15	820	32	80
700	3	28	840	37	86
720	5	40	860	40	88

TABLE AII. Spectral reflectance values (percent) for Tan 380 and Tan 686

Wavelength, Nanometers (nm)	Reflectance Min.	Max.
700	45	65
720	45	65
740	45	65
760	45	65
780	45	65
800	45	65
820	45	65
840	45	65
860	45	65"

PAGE 7

Table I, after the heading "Inhibitor": Add "2/"

Add the following new footnote:

"2/ When desert or light shades are specified, inhibitor (d) or (e) shall not be used."

PAGE 12

Table II, under "Shrinkage in laundering": Add the following:

"Spectral reflectance ----- 3.6.2 ----- 4.5.2"

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:31Z
Check the source to verify that this is the current version before use.

PAGE 17

Add the following new paragraph:

"4.5.2 Spectral reflectance measurements in the visible/near infrared. Spectral reflectance data shall be obtained from 600 to 860 nm for the olive green and olive drab shades and from 700 to 860 nm for the Tan 380 and Tan 686, at 20 nm intervals on a spectrophotometer (see 6.9) relative to a barium sulfate standard, the preferred white reference standard. Other white reference materials may be used, provided they are calibrated to aboslute white; e.g. Halon, magnesium oxide, or vitrolite tiles (see 6.10). The spectral band width at 860 nm shall be less than 26 nm. Reflectance measurements shall be made by either the monochromatic or polychromatic mode of operation. When the polychromatic mode is used, the spectrophotometer shall operate with the specimen diffusely illuminated with full emission of a continuous source that simulates in the visible spectrum either CIE Source A or CIE Source D65. The specimen shall be measured as a single layer backed with two layers of the same fabric and shade. Readings shall be taken on a minimum of two different areas and the data averaged. The specimen shall be viewed at an angle no greater than 10 degrees from the normal. Photometric accuracy of the spectrophotometer shall be within 1 percent and the wavelength accuracy shall be within 2 nm. When the measured reflectance values for any color at four or more of the listed wavelengths do not meet the limits specified in tables AI and AII, it shall be considered a test failure."

PAGE 18

Add the following paragraphs:

"6.9 Spectrophotometers. Suitable spectrophotometers for measuring spectral reflectance in the visible/near infrared are the Diano Hardy, Diano Match Scan, Hunter D54P-IR, and Macbeth 1500 with IR option.

6.10 White standard. Barium sulfate of suitable quality for use as a white reference standard is available from the Eastman Kodak Company. The same source has available, magnesium reagent (ribbon) and Halon. Suitable tiles can be obtained from the National Institute of Standards and Technology or the instrument manufacturers."

Preparing activity:

Army - GL

(Project 8305-A417)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:31Z
Check the source to verify that this is the current version before use.